

WASP-35b

R-1005

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_260202 · created 2026-07-30T01:58:17+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461073.7019 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.048
Transit depth (Rp/Rs)^2	2.71 +/- 1.58 [%]
Orbital Inclination (inc)	88.71 +/- 1.91
Airmass coefficient 1 (a1)	1.031 +/- 0.042
Airmass coefficient 2 (a2)	-0.014 +/- 0.03
Scatter in the residuals of the lightcurve fit is	4.07 %
Best Comparison Star	#2 - [440, 118]
Optimal Aperture	2.87
Optimal Annulus	4.89
Transit Duration (day)	0.0946 +/- 0.0076

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.165 $\pm$ 0.048 vs 0.1241 (deviation 0.77 σ)
Duration fit vs published	0.0946 d vs 0.1317 d (deviation 4.43 σ)
Mid-time O-C	-11.1 min
Depth SNR	3.1
Residual scatter	4.07 %

Light curve

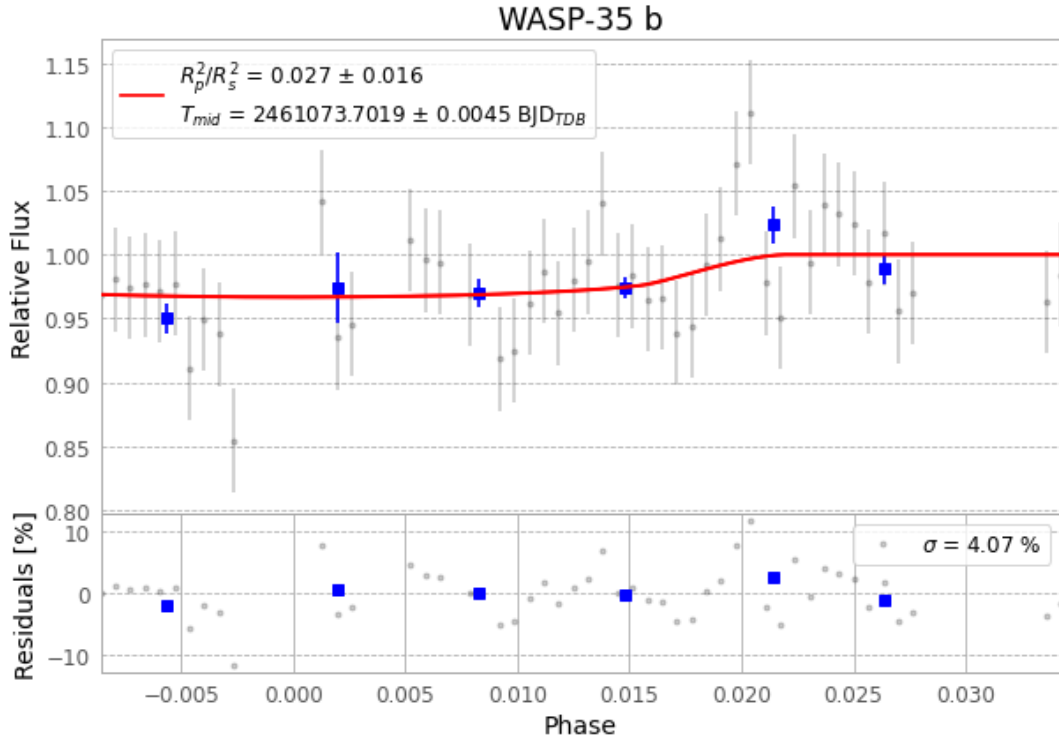


Figure 1 — FinalLightCurve_WASP-35 b_2026-02-01.png (SHA-256 302a0c99900b4c48...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [320, 171], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 30bbb43d6c79f9c7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

70 FITS frames (46 MB), WASP-35260202040616.FITS → WASP-35260202073315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-260202.log (ce78caa4d1fe...), FinalLightCurve_WASP-35 b_2026-02-01.png (302a0c99900b...), FinalLightCurve_WASP-35 b_2026-02-01.csv (1f36503e4a9f...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 01:58Z.